5.1 Communication and homeostasis

5.1.1 Communication and homeostasis

Organisms use both chemical and electrical systems to monitor and respond to any deviation from the body's steady state.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the need for communication systems in multicellular organisms	To include the need for animals and plants to respond to changes in the internal and external environment and to coordinate the activities of different organs.
(b)	the communication between cells by cell signalling	To include signalling between adjacent cells and signalling between distant cells.
(c)	the principles of homeostasis	To include the differences between receptors and effectors, and the differences between negative feedback and positive feedback. HSW8
(d)	the physiological and behavioural responses involved in temperature control in ectotherms and endotherms.	To include, • endotherms – peripheral temperature receptors, the role of the hypothalamus and effectors in skin and muscles; behavioural responses • ectotherms – behavioural responses. An opportunity to monitor physiological functions in ectotherms and/or endotherms. **PAG11** HSW2